Import necessary libaries

In [1]:
```
# Data manipulation
import pandas as pd
import numpy as np

# Visualization
import matplotlib.pyplot as plt
import seaborn as sns
```

In [ ]:

Load the dataset into notebook

In [2]:
```
df = pd.read_csv('customer_churn_dataset.csv')
```

In [3]:
```
# Preview of the Dataset
df.head()
```

Out[3]:

	Age	Gender	Tenure_Months	Monthly_Charges	Contract_Type	Internet_Service	Support_Calls	Payment_Method	Total_Charges	Churn
0	56	Female	65	84.01	Month-to-month	DSL	1	Card	5460.65	No
1	69	Male	54	46.97	Month-to-month	Fiber	4	Bank Transfer	2536.38	Yes
2	46	Female	52	86.86	Month-to-month	NaN	2	Bank Transfer	4516.72	No
3	32	Female	3	87.12	One year	NaN	0	Cash	261.36	No
4	60	Male	41	76.07	Month-to-month	Fiber	5	Cash	3118.87	Yes

In [ ]:

In [4]:
```
# Dataset Dimensions
df.shape
```

Out[4]: (1500, 10)

In [5]:
```
df.columns.tolist
```

Out[5]:
```
<bound method IndexOpsMixin.tolist of Index(['Age', 'Gender', 'Tenure_Months', 'Monthly_Charges', 'Contract_Type',
       'Internet_Service', 'Support_Calls', 'Payment_Method', 'Total_Charges',
       'Churn'],
      dtype='object')>
```

In [6]:
```
# Dataset Structure and Data Types
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1500 entries, 0 to 1499
Data columns (total 10 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Age               1500 non-null   int64
 1   Gender            1500 non-null   object
 2   Tenure_Months     1500 non-null   int64
 3   Monthly_Charges   1500 non-null   float64
 4   Contract_Type     1500 non-null   object
 5   Internet_Service  1366 non-null   object
 6   Support_Calls     1500 non-null   int64
 7   Payment_Method    1500 non-null   object
 8   Total_Charges     1500 non-null   float64
 9   Churn             1500 non-null   object
dtypes: float64(2), int64(3), object(5)
memory usage: 117.3+ KB
```

In [7]:
```
# Summary Statistics of Numerical Features
df.describe()
```

Out[7]:

	Age	Tenure_Months	Monthly_Charges	Support_Calls	Total_Charges
count	1500.000000	1500.000000	1500.000000	1500.000000	1500.000000
mean	43.732000	35.904667	69.890627	2.479333	2499.317740
std	15.071771	20.299022	17.197149	1.686654	1570.378869
min	18.000000	1.000000	40.010000	0.000000	45.260000
25%	31.000000	19.000000	55.215000	1.000000	1188.220000
50%	44.000000	36.000000	70.120000	2.000000	2389.485000
75%	56.000000	53.250000	84.870000	4.000000	3554.572500
max	69.000000	71.000000	99.860000	5.000000	7075.860000

In [ ]:

In [8]:
```
# Check for Duplicate Records
df.duplicated().sum()
```

Out[8]:
```
np.int64(0)
```

In [9]:
```
# Calculate percentage of missing values per column
missing_percentage = (df.isnull().sum() / len(df)) * 100
missing_percentage.sort_values(ascending=False)
```

Out[9]:
```
Internet_Service    8.933333
Age                 0.000000
Gender              0.000000
Tenure_Months       0.000000
Monthly_Charges     0.000000
Contract_Type       0.000000
Support_Calls       0.000000
Payment_Method      0.000000
Total_Charges       0.000000
Churn               0.000000
dtype: float64
```

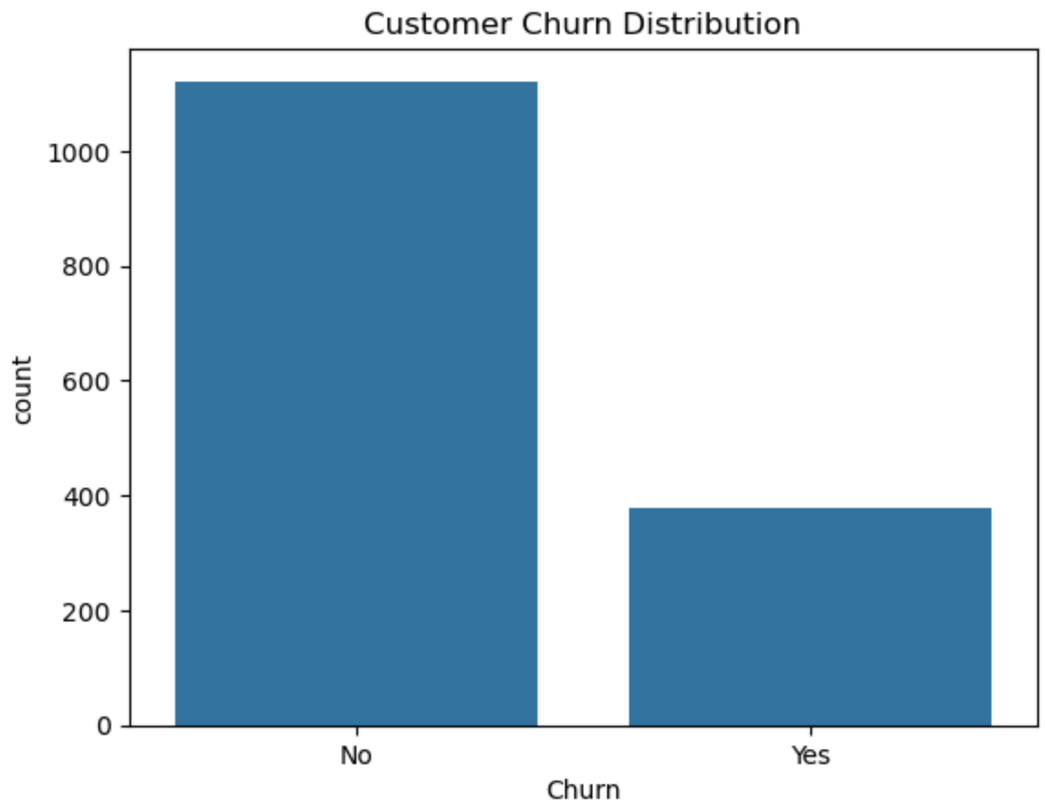
In [10]:
```
df['Internet_Service'] = df['Internet_Service'].fillna(df['Internet_Service'].mode()[0])
```

In [11]:
```
missing_percentage = (df.isnull().sum() / len(df)) * 100
missing_percentage.sort_values(ascending=False)
```

Out[11]:
```
Age                 0.0
Gender              0.0
Tenure_Months       0.0
Monthly_Charges     0.0
Contract_Type       0.0
Internet_Service    0.0
Support_Calls       0.0
Payment_Method      0.0
Total_Charges       0.0
Churn               0.0
dtype: float64
```

Exploratory Data Analysis (EDA)

In [12]:
```
# Churn distribution
sns.countplot(x="Churn", data=df)
plt.title("Customer Churn Distribution")
plt.show()
```



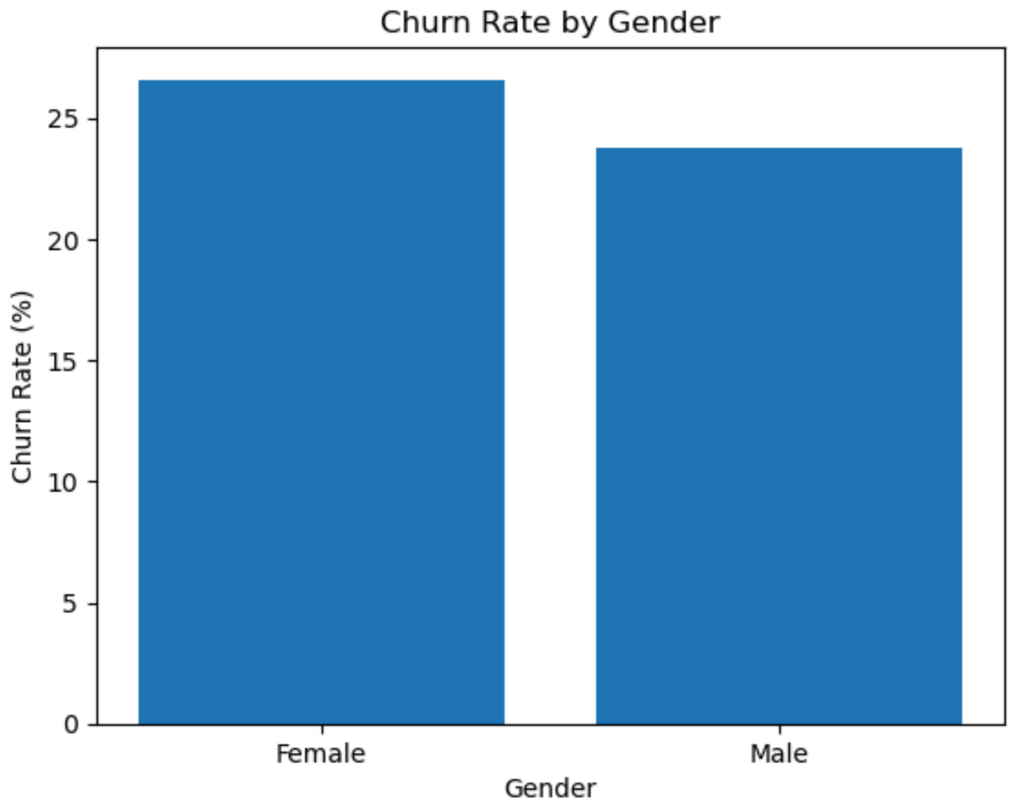
In [13]:
```
# Calculate churn rate by gender
churn_by_gender = (df.groupby("Gender")["Churn"].apply(lambda x: (x == "Yes").mean() * 100).reset_index(name="Churn_Rate (%)"))
churn_by_gender
```

Out[13]:

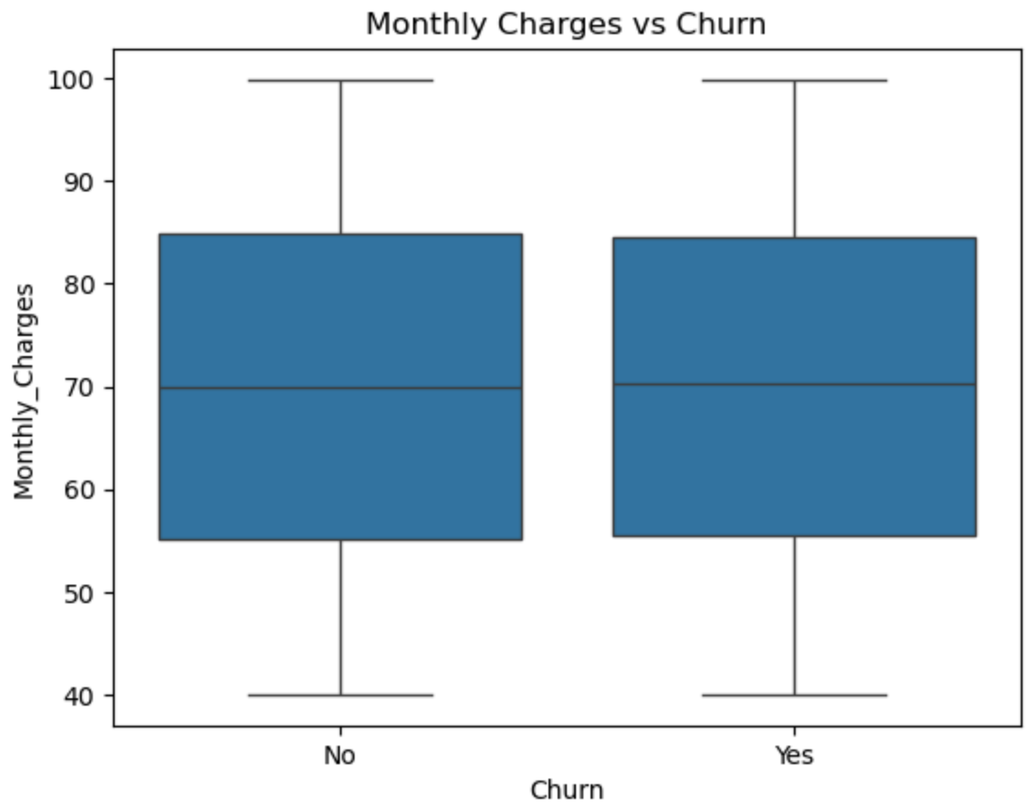
	Gender	Churn_Rate (%)
0	Female	26.587302
1	Male	23.790323

In [14]:
```
plt.figure()
plt.bar(churn_by_gender["Gender"], churn_by_gender["Churn_Rate (%)"])
plt.title("Churn Rate by Gender")
```

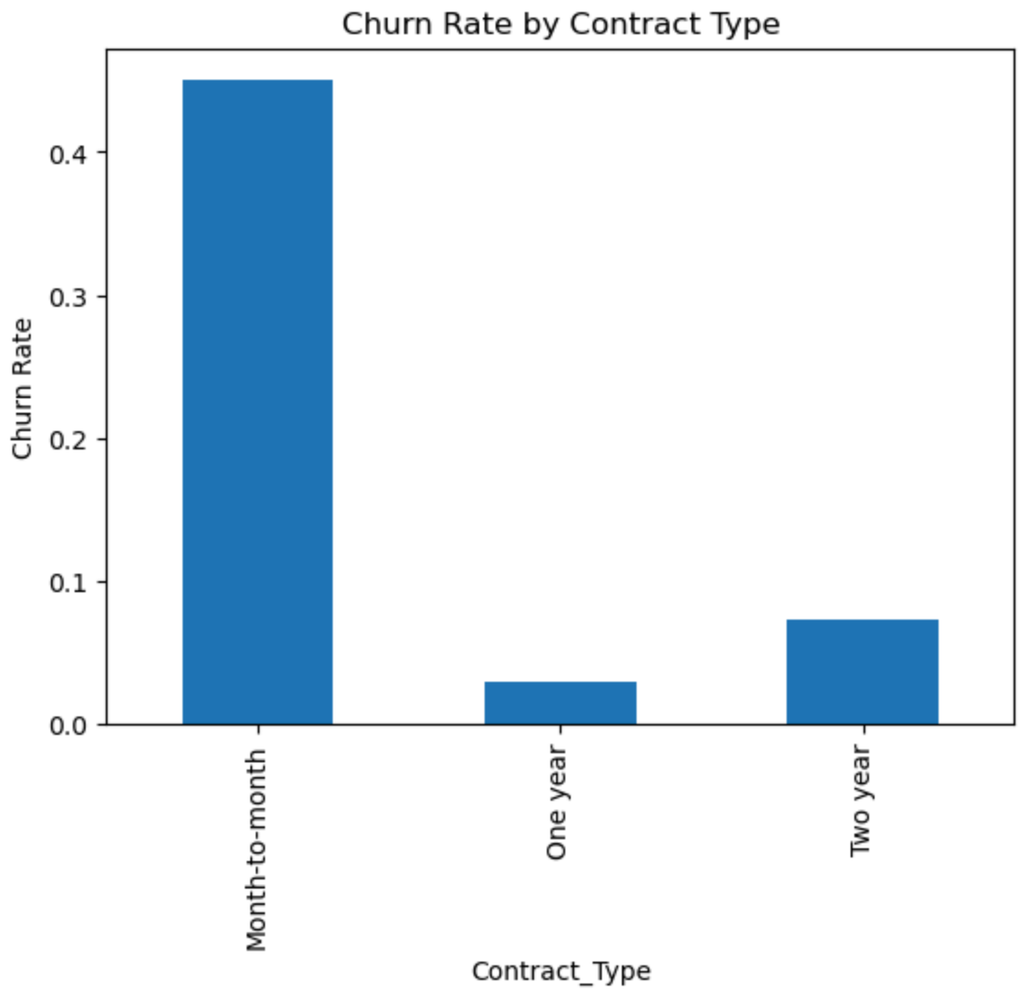
```
plt.xlabel("Gender")
plt.ylabel("Churn Rate (%)")
plt.show()
```



In [15]:
```
# Monthly Charges vs Churn
sns.boxplot(x="Churn", y="Monthly_Charges", data=df)
plt.title("Monthly Charges vs Churn")
plt.show()
```



In [16]:
```
# Churn rate by Contract Type
contract_churn = df.groupby("Contract_Type")["Churn"].apply(lambda x: (x=="Yes").mean())
contract_churn.plot(kind="bar", title="Churn Rate by Contract Type")
plt.ylabel("Churn Rate")
plt.show()
```



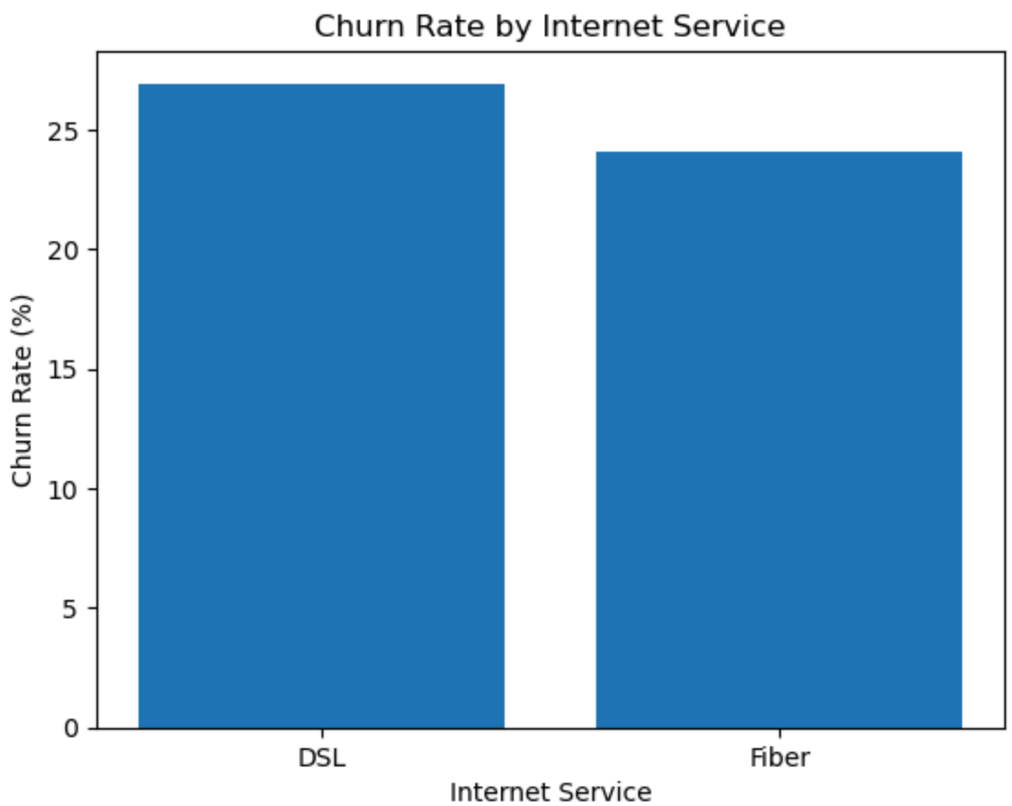
In [17]:
```
# Calculate churn rate by internet service
churn_by_internet = (df.groupby("Internet_Service")["Churn"].apply(lambda x: (x == "Yes").mean() * 100).reset_index(name="Churn_Rate (%)"))
churn_by_internet
```

Out[17]:

	Internet_Service	Churn_Rate (%)
0	DSL	26.943005
1	Fiber	24.104235

In [18]:

```
plt.figure()
plt.bar(churn_by_internet["Internet_Service"], churn_by_internet["Churn_Rate (%)"])
plt.title("Churn Rate by Internet Service")
plt.xlabel("Internet Service")
plt.ylabel("Churn Rate (%)")
plt.show()
```

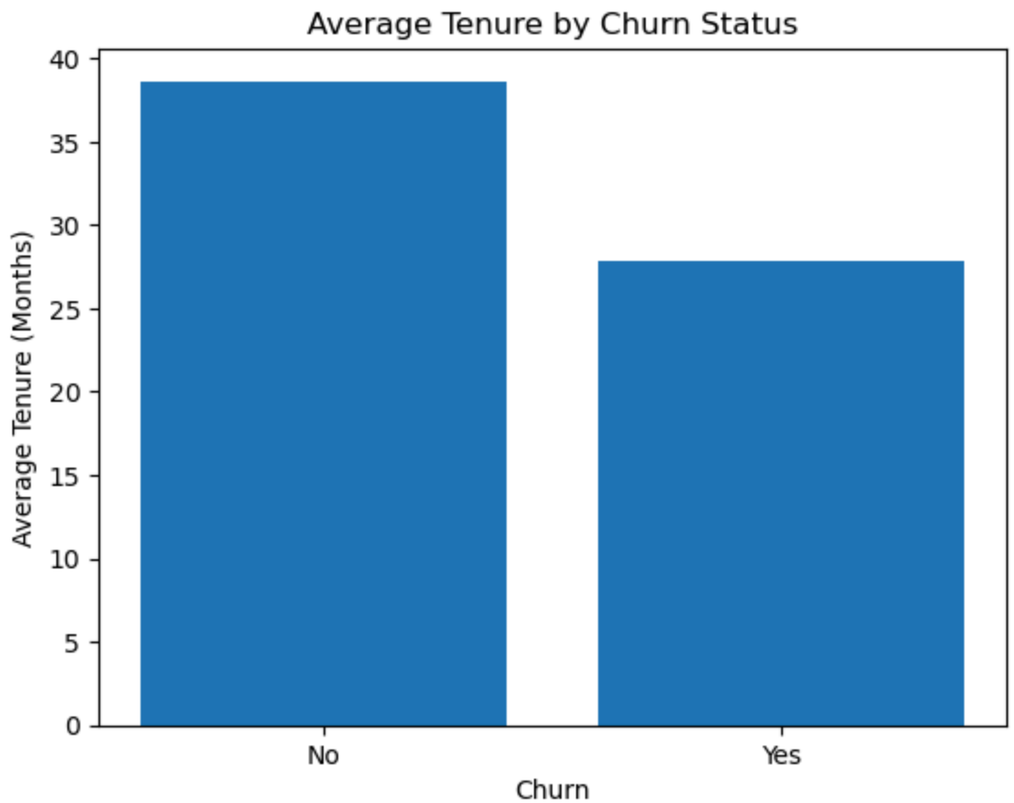


In [19]:
```
# Average tenure by churn status
tenure_churn = df.groupby("Churn")["Tenure_Months"].mean().reset_index()
tenure_churn
```

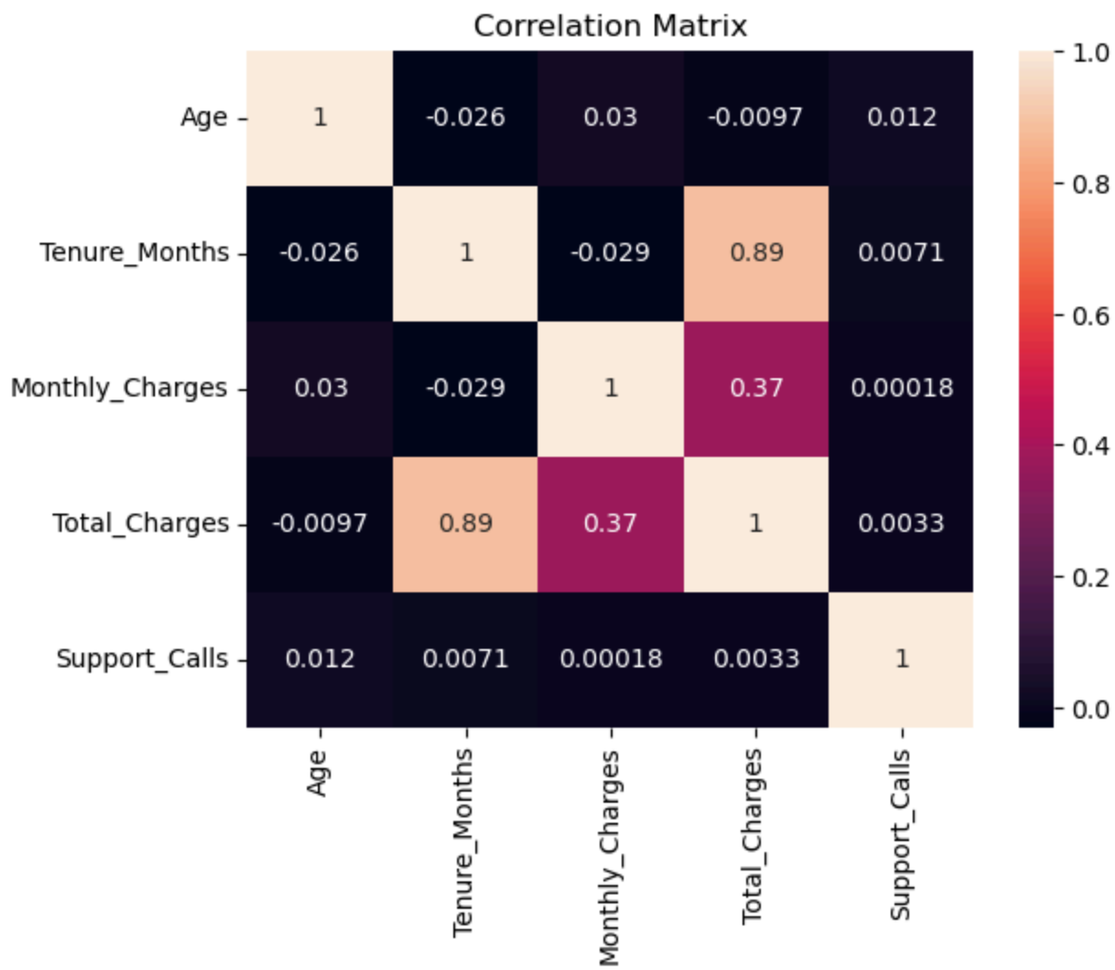
Out[19]:

	Churn	Tenure_Months
0	No	38.632799
1	Yes	27.806878

In [20]:
```
plt.figure()
plt.bar(tenure_churn["Churn"], tenure_churn["Tenure_Months"])
plt.title("Average Tenure by Churn Status")
plt.xlabel("Churn")
plt.ylabel("Average Tenure (Months)")
plt.show()
```



In [21]:
```
# Correlation heatmap for numeric features
sns.heatmap(df[["Age", "Tenure_Months", "Monthly_Charges", "Total_Charges", "Support_Calls"]].corr(), annot=True)
plt.title("Correlation Matrix")
plt.show()
```



Data Preprocessing

- Preprocessing and ML

In [22]:
```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
```

```
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import confusion_matrix, classification_report, roc_auc_score
import joblib

# Display options
pd.set_option("display.max_columns", None)
```

In [23]:
```
# Define features and target
X = df.drop("Churn", axis=1)
y = df["Churn"].map({"Yes": 1, "No": 0})
```

In [ ]:

In [ ]:

In [24]:
```
# Numerical and categorical features
num_features = ["Age", "Tenure_Months", "Monthly_Charges", "Total_Charges"]
cat_features = ["Contract_Type", "Internet_Service", "Payment_Method"]
```

In [25]:
```
# Column transformer
preprocessor = ColumnTransformer([
    ("num", StandardScaler(), num_features),
    ("cat", OneHotEncoder(drop="first"), cat_features)
])
```

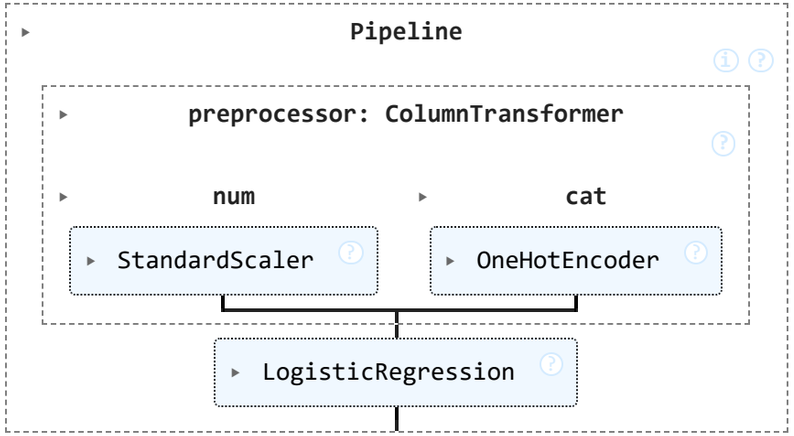
In [26]:
```
# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, stratify=y, random_state=42)
```

Build Machine Learning Models

- Logistic Regression

In [27]:
```
log_model = Pipeline([("preprocessor", preprocessor), ("classifier", LogisticRegression(max_iter=1000))])
log_model.fit(X_train, y_train)
```

Out[27]:



In [28]:

```
# Predict labels on test set
y_pred_log = log_model.predict(X_test)

# Predict probabilities on test set
y_prob_log = log_model.predict_proba(X_test)[:, 1]
```

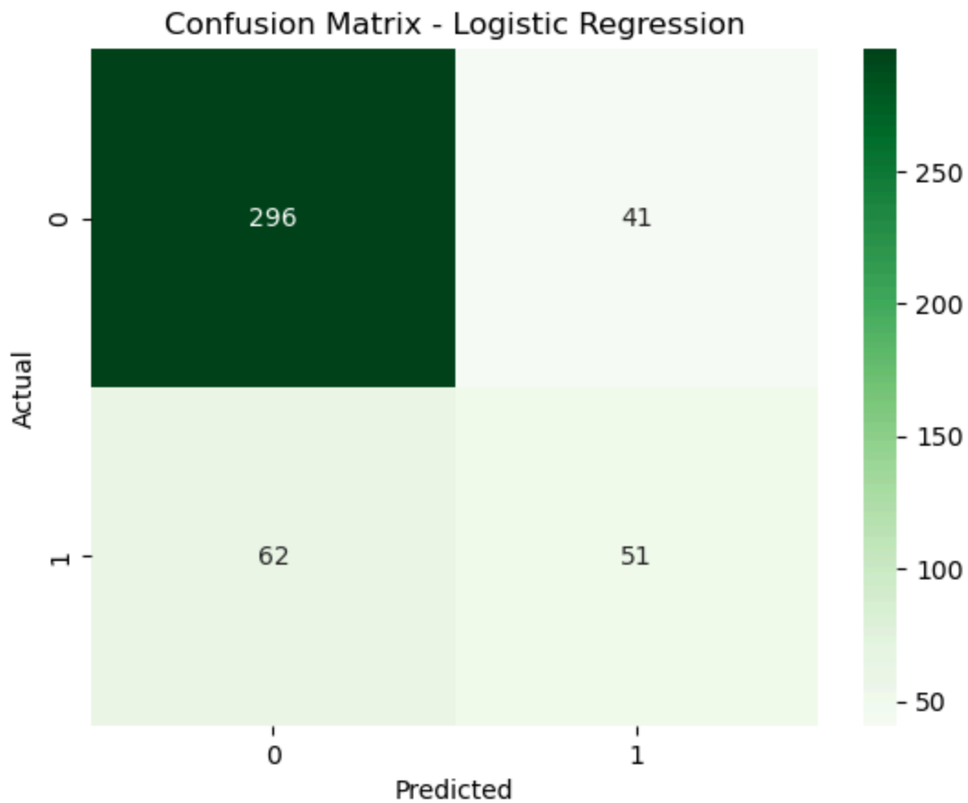
In [29]:

```
cm_log = confusion_matrix(y_test, y_pred_log)
print("Confusion Matrix - Logistic Regression:\n", cm_log)
```

```
Confusion Matrix - Logistic Regression:
 [[296  41]
 [ 62  51]]
```

In [30]:

```
sns.heatmap(cm_log, annot=True, fmt="d", cmap="Greens")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix - Logistic Regression")
plt.show()
```



In [ ]:

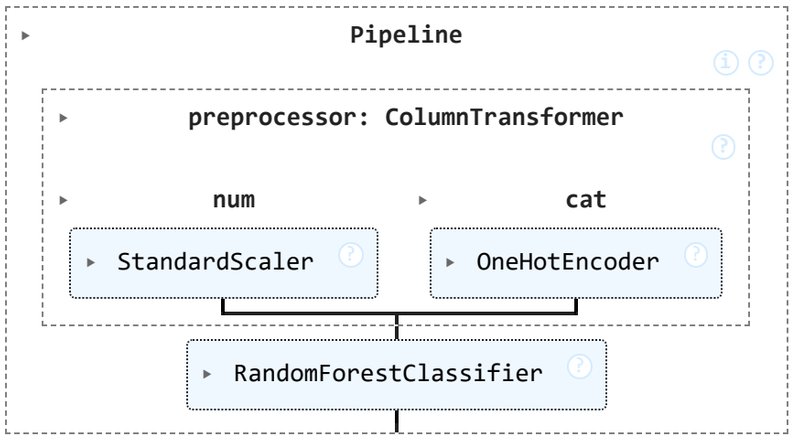
In [31]:
```
report_log = classification_report(y_test, y_pred_log, target_names=["No Churn", "Churn"])
print("Classification Report - Logistic Regression:\n")
print(report_log)
```

```
Classification Report - Logistic Regression:

              precision    recall  f1-score   support

    No Churn       0.83      0.88      0.85       337
       Churn       0.55      0.45      0.50       113

    accuracy                           0.77       450
   macro avg       0.69      0.66      0.67       450
weighted avg       0.76      0.77      0.76       450
```

Random Forest

In [32]:
```
rf_model = Pipeline([("preprocessor", preprocessor), ("classifier", RandomForestClassifier(n_estimators=200, random_state=42))])
rf_model.fit(X_train, y_train)
```

Out[32]:



Model Evaluation

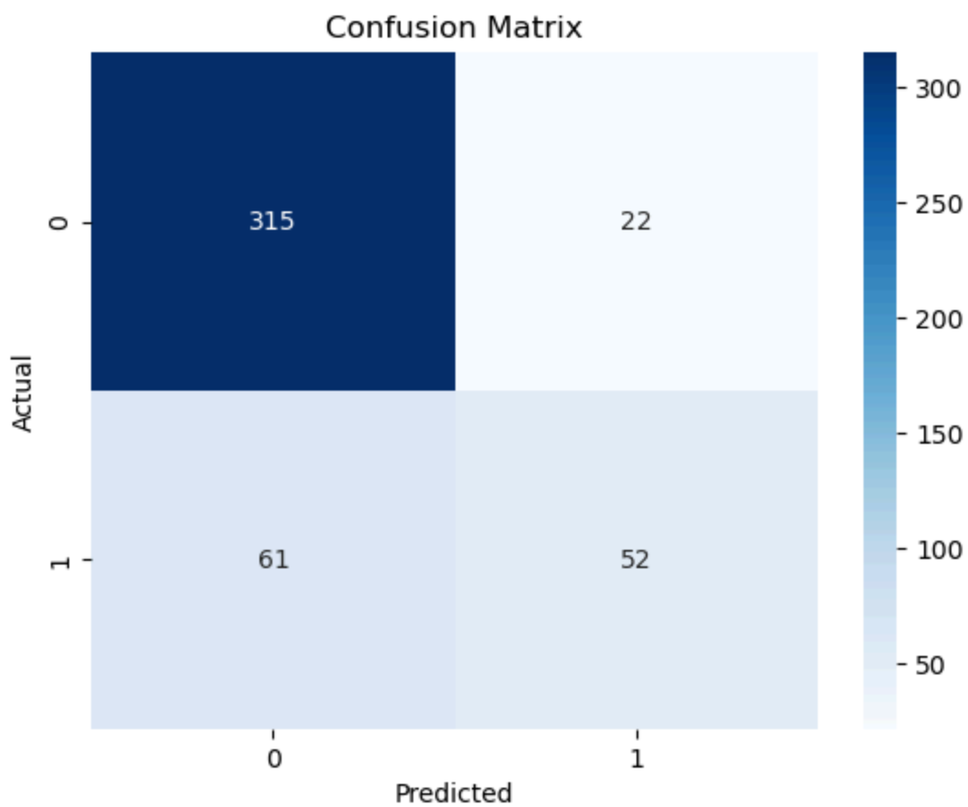
In [33]:
```
# Predict on test set
y_pred = rf_model.predict(X_test)
y_prob = rf_model.predict_proba(X_test)[:,1]
```

In [34]:
```
# Confusion matrix
cm = confusion_matrix(y_test, y_pred)
```

```
print("Confusion Matrix:\n", cm)
```

```
Confusion Matrix:
 [[315  22]
 [ 61  52]]
```

In [35]:
```
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



In [36]:
```
# Classification report
print("Classification Report - Random Classifier:\n", classification_report(y_test, y_pred))
```

```
Classification Report - Random Classifier:
              precision    recall  f1-score   support

           0       0.84      0.93      0.88       337
           1       0.70      0.46      0.56       113

    accuracy                           0.82       450
   macro avg       0.77      0.70      0.72       450
weighted avg       0.80      0.82      0.80       450
```

Feature Importance

In [37]:
```
# Extract feature importances
rf_model_steps = rf_model.named_steps["classifier"]
feature_names = rf_model.named_steps["preprocessor"].get_feature_names_out()
importances = rf_model_steps.feature_importances_
```

In [38]:
```
fi = pd.DataFrame({"Feature": feature_names, "Importance": importances}).sort_values(by="Importance", ascending=False)
fi.head(10)
```

Out[38]:

	Feature	Importance
1	num__Tenure_Months	0.221088
3	num__Total_Charges	0.206529
2	num__Monthly_Charges	0.164507
4	cat__Contract_Type_One year	0.135811
0	num__Age	0.131635
5	cat__Contract_Type_Two year	0.086162
6	cat__Internet_Service_Fiber	0.019726
8	cat__Payment_Method_Cash	0.017284
7	cat__Payment_Method_Card	0.017258

Model for Deployment

In [39]:
```
# Save the trained model
joblib.dump(rf_model, "churn_model.pkl")
```

Out[39]:
```
['churn_model.pkl']
```

In [ ]:
```
streamlit run customer_churn_app.py
```

In [ ]: